PHYS 7502 – Homework 5, Problem 5 (Extended Treatment)

Why Calorimeters Win at High Energy and Trackers Win at Low Energy

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References: Mann, Ch. 8; Thomson, Modern Particle Physics*, Ch. 1; Particle Data Group (PDG)* Passage of Particles through Matter *review.*

Problem Statement

Explain why, *in terms of precision*, calorimeters are the detectors of choice for high-energy particles, while charged-particle tracking is preferred below 5–10 GeV.

What the question is really asking. At a collider or fixed-target experiment, you have (at least) two independent ways to determine how much momentum or energy a charged particle is carrying:

(i) Let it fly undisturbed through a magnetic field and measure the curvature of its trajectory – a *tracker*.

(ii) Stop it in a dense medium and measure the total signal deposited as it showers and ionizes – a *calorimeter*.

Both methods have a characteristic fractional uncertainty (σ_p/p for a tracker, σ_E/E for a calorimeter) that depends on the particle's energy – but the two functions scale in *opposite directions*. The question asks us to (a) derive those two scalings from first principles, (b) identify the energy at which they cross, and (c) give the physical reason the scalings go the way they do.

Deliverables

- Distinguish the two observables being measured (p vs. E).
- Derive the tracker momentum resolution from the Lorentz force and track sagitta, identifying the $\sigma_p/p \propto p$ scaling.
- Derive the calorimeter energy resolution from shower-particle counting statistics, identifying the $\sigma_E/E \propto 1/\sqrt{E}$ scaling.
- Solve for the crossover energy $E_\times$ where the two resolutions are equal.
- Justify why $E_\times$ sits around 5–10 GeV for the charged-hadron / HCAL case and 30–50 GeV for the EM case.
- Note the non-precision reasons calorimetry still wins at high energy (neutrals, jets, trigger speed).

1 Step 1 – Two different things are being measured

A tracker reports *momentum*; a calorimeter reports *energy*. For a highly relativistic particle these are numerically almost equal ($E \simeq pc$), which is why one often casually equates the two. But they are not the same observable, and the physical process by which each is extracted is what drives their opposite precision trends.

Momentum vs. energy: a crucial distinction

- **Tracker – momentum measurement.** The particle is left alive; one infers $\vec{p}$ from the *shape* of its trajectory in a known magnetic field $\vec{B}$. The measurement is *geometric and indirect*: the detector never "knows" the energy, only the positions of a handful of hits.
- **Calorimeter – energy measurement.** The particle is destroyed (absorbed); one infers E from the *total signal* produced as the particle's energy is cascaded down into a shower of many low-energy secondaries. The measurement is *statistical and direct*: the detector sums up many small contributions that together account for all the energy.

Because these are fundamentally different physical processes – geometry-limited position fitting vs. counting-statistics on shower quanta – the way their uncertainties scale with the particle's energy is also fundamentally different.

With this framing, the two derivations below reduce to: "how small a curvature can I still measure?" (tracker) and "how many independent shower quanta am I averaging over?" (calorimeter).

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2 Step 2 – The tracker: geometry sets the precision

2.1 How a tracker measures momentum

A charged particle of charge q moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$ feels the Lorentz force

$$\vec{F} = q\,\vec{v} \times \vec{B}. \tag{1}$$

If $\vec{B}$ is uniform and perpendicular to $\vec{v}$, the force provides the centripetal acceleration for a circular arc of radius R:

$$\frac{pv}{R} \;=\; qvB \qquad \Longrightarrow \qquad p \;=\; qBR. \tag{2}$$

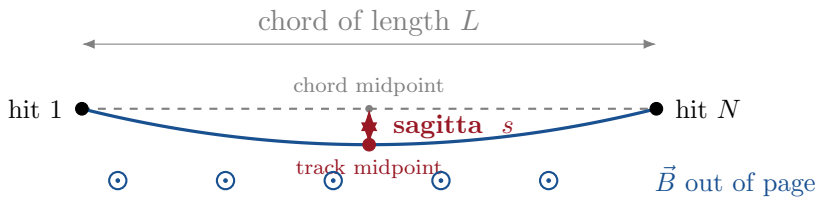
Putting in SI/natural-unit constants ($c = 3 \times 10^8\,\mathrm{m/s}$, convert momentum from kg·m/s to GeV/c):

$$\boxed{p_T\,[\mathrm{GeV}/c] \;=\; 0.2998\,B\,[\mathrm{T}]\,R\,[\mathrm{m}]} \tag{3}$$

The useful number 0.3 is worth committing to memory. It says: a $1\,\mathrm{GeV}/c$ charged particle in a 1 T field has a radius of curvature of about $\frac{1}{0.3} \approx 3.33\,\mathrm{m}$.

2.2 Why resolution degrades at high momentum: the sagitta

The radius of curvature R is huge compared to the size of a real tracker, so we never see a full circle – only a short, gently curved arc. The practical quantity a tracker actually measures is the *sagitta* s: the perpendicular distance between the midpoint of the track and the chord connecting its two endpoints.



From basic circle geometry, for a chord of length L much smaller than R, the sagitta is

$$s \;=\; R\left(1-\cos\tfrac{\theta}{2}\right) \;\approx\; \frac{R\,\theta^2}{8} \;=\; \frac{L^2}{8R}, \tag{4}$$

where I used $\theta \approx L/R$ and $\cos(\theta/2) \approx 1 - \theta^2/8$ for small θ. Combining with Eq. (3):

$$\boxed{s \;=\; \frac{0.3\,BL^2}{8\,p_T}} \tag{5}$$

The central punchline of tracker precision

The sagitta is *inversely proportional* to momentum: a higher-momentum particle makes a *smaller* sagitta, because it bends less. A tracker can only measure a finite minimum displacement – set by the position resolution σ_x of each hit. So as p_T grows, the sagitta s shrinks toward σ_x, and the fractional uncertainty σ_s/s blows up.
Quantitatively: take $B = 2\,\mathrm{T}$, $L = 1\,\mathrm{m}$, $\sigma_x = 10\,\mu\mathrm{m}$. A $1\,\mathrm{GeV}/c$ track has $s \approx 75\,\mathrm{mm}$: huge, trivially measured. A $100\,\mathrm{GeV}/c$ track has $s \approx 0.75\,\mathrm{mm}$: still fine. A $1\,\mathrm{TeV}/c$ track has $s \approx 75\,\mu\mathrm{m}$: only about $7\sigma_x$ – barely distinguishable from a straight line.

2.3 Gluckstern's formula: the tracker resolution with N hits

Differentiating Eq. (5) at fixed B, L:

$$\frac{\sigma_{p_T}}{p_T} \;=\; \frac{\sigma_s}{s}, \tag{6}$$

since $s \propto 1/p_T$, so $|d\ln s| = |d\ln p_T|$. With N equally spaced position measurements along the track, each with individual precision σ_x, a quadratic fit to the trajectory gives a sagitta uncertainty of

$$\sigma_s \;=\; \sigma_x \sqrt{\frac{720}{N+4}}\,\frac{1}{8}, \tag{7}$$

which is the *Gluckstern formula* (1963). Substituting into the previous line:

$$\boxed{\left(\frac{\sigma_{p_T}}{p_T}\right)_{\!\mathrm{meas}} \;=\; \frac{\sigma_x\,p_T}{0.3\,BL^2}\sqrt{\frac{720}{N+4}}\,,} \tag{8}$$

where p_T is in GeV/c, B in T, L in m, and σ_x in m.

Interpreting every factor

- σ_x (numerator): finer position measurements ⇒ better momentum resolution. Silicon strips at $25\,\mu\mathrm{m}$ beat a gas drift chamber at $200\,\mu\mathrm{m}$.
- p_T (numerator): *this is the crucial factor.* Higher momentum means smaller sagitta, hence

worse fractional resolution. The resolution *worsens linearly with momentum.*

- B (denominator): stronger field bends the track more, larger sagitta, better resolution. This is why LHC detectors use $B \sim 2$–$4\,\mathrm{T}$ solenoids.
- L^2 (denominator): longer tracker gives *quadratically* better resolution. This is a huge lever arm and is why the CMS tracker extends to $r \approx 1\,\mathrm{m}$.
- $\sqrt{720/(N+4)}$: more hits along the track fit the parabola better. This factor is only a mild ($\sim N^{-1/2}$) improvement compared to the L^2 lever.

2.4 Multiple scattering: a floor at low momentum

Equation (8) naively predicts that low-p_T tracks have *perfect* momentum resolution. They do not, because a second effect kicks in at low momentum: *multiple Coulomb scattering* off the atomic nuclei of the tracker material. Each scatter randomly kinks the trajectory by a small angle; the rms kink after traversing a thickness x/X_0 of a medium with radiation length X_0 is

$$\theta_0 \;=\; \frac{13.6\,\mathrm{MeV}}{\beta c\, p_T}\sqrt{\frac{x}{X_0}}\left(1 + 0.038\ln\frac{x}{X_0}\right). \tag{9}$$

These kinks add an extra uncertainty to the reconstructed sagitta that is independent of p_T in the relative sense (both s and σ_s^{MS} scale as $1/p_T$, so their ratio is constant). The resulting contribution is

$$\left(\frac{\sigma_{p_T}}{p_T}\right)_{\mathrm{MS}} \;\approx\; \frac{0.045}{BL}\sqrt{\frac{x/X_0}{\beta\cos\lambda}}, \tag{10}$$

independent of p_T for $\beta \approx 1$.

The two contributions add in quadrature:

$$\boxed{\left(\frac{\sigma_{p_T}}{p_T}\right)^2 \;=\; \underbrace{a_{\mathrm{MS}}^2}_{\text{low-}p_T\text{ floor}} \;+\; \underbrace{(b\,p_T)^2}_{\text{high-}p_T\text{ tail}}\,,} \tag{11}$$

with typical LHC values $a_{\mathrm{MS}} \approx 0.3$–$0.5\,\%$ and $b \approx 0.02$–$0.05\,\%/\mathrm{GeV}$. Below a few GeV the first term dominates and the resolution is nearly constant; above it, the second term dominates and the resolution grows linearly.

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3 Step 3 – The calorimeter: statistics sets the precision

3.1 How a calorimeter measures energy

Place a GeV-scale electron in front of a block of dense material. It will:

1. Emit a high-energy bremsstrahlung photon within one radiation length X_0.
2. That photon will pair-produce within roughly $\frac{9}{7}X_0$, giving a new e^+e^- pair each carrying half the photon's energy.
3. Each of those daughter electrons and positrons repeats step 1, producing more photons, more pairs…

This chain is an *electromagnetic shower*. In a simple Heitler-style cartoon, after n generations the shower contains 2^n particles each with energy $E/2^n$. The shower stops multiplying once per-particle energy drops below the critical energy E_c (where ionization loss starts to dominate over bremsstrahlung). The total number of particles in the shower at maximum is therefore

$$N_{\max} \sim \frac{E}{E_c} \propto E. \tag{12}$$

A hadronic shower is messier (nuclear interactions, $\pi^0 \to \gamma\gamma$ feeding an EM component, neutrons wandering off, $\sim 30\%$ energy lost to nuclear break-up), but the same basic scaling $N \propto E$ holds because each GeV of incoming energy produces roughly a fixed number of charged secondaries.

Why $N \propto E$ is what matters

Equation (12) says that *the shower size grows linearly with the incident energy.* Every tracked shower particle deposits an amount of signal (ionization charge, scintillation photons, Cherenkov photons) that is roughly fixed. So the total collected signal is

$$S = \eta N \propto E,$$

where η is a material- and geometry-dependent collection efficiency. This is the core property that makes a calorimeter *linear* in energy: the pulse height is proportional to E, independent of the details of the shower.

3.2 Why Poisson statistics gives the $1/\sqrt{E}$ scaling

The shower is built up from N approximately independent branchings, and the signal is a sum of many small contributions, each a sample from some distribution. Whenever one counts independent discrete events, the central limit theorem and Poisson statistics give

$$\sigma_N = \sqrt{N} \qquad \Longrightarrow \qquad \frac{\sigma_N}{N} = \frac{1}{\sqrt{N}}. \tag{13}$$

Since $S \propto N$, the relative uncertainty of the measured signal is the same:

$$\frac{\sigma_S}{S} = \frac{1}{\sqrt{N}} \propto \frac{1}{\sqrt{E}}. \tag{14}$$

That is the entire argument for the $1/\sqrt{E}$ scaling. It comes directly from the fact that the shower is a sum of many Poisson samples, nothing more.

3.3 The three-term calorimeter resolution formula

In practice one adds two smaller contributions to the stochastic term:

$$\boxed{\frac{\sigma_E}{E} = \underbrace{\frac{a}{\sqrt{E/\mathrm{GeV}}}}_{\text{stochastic}} \oplus \underbrace{b}_{\text{constant}} \oplus \underbrace{\frac{c}{E/\mathrm{GeV}}}_{\text{noise}},} \tag{15}$$

where $\oplus$ means quadrature sum. Physical meaning of each term:

The three terms of Eq. (15)

- **Stochastic term $a/\sqrt{E}$:** counting statistics on the shower, as derived above. Dominates at low energy. Typical a: 3–10% for EM calorimeters, 50–100% for hadronic calorimeters (bigger because hadronic showers have lower N per unit energy – much of the shower energy goes into neutrons and nuclear break-up, not into signal-producing charged particles).
- **Constant term b:** calibration imperfections, inter-channel intercalibration, longitudinal shower leakage, detector nonuniformity. These are systematic and do not benefit from higher statistics, so they survive to high energy and set the asymptotic floor on resolution. Typical b: $< 1\%$ for modern EM calorimeters, a few percent for hadronic.
- **Noise term c/E:** electronic noise, pileup from other simultaneous collisions. A fixed amount of noise per channel becomes relatively less important at higher shower energies, so this term is steeply suppressed at high E. Typical c: tens of MeV to a GeV.

At the energies that matter for a crossover with tracking, *only the stochastic term matters*, and we simplify to $\sigma_E/E \approx a/\sqrt{E}$.

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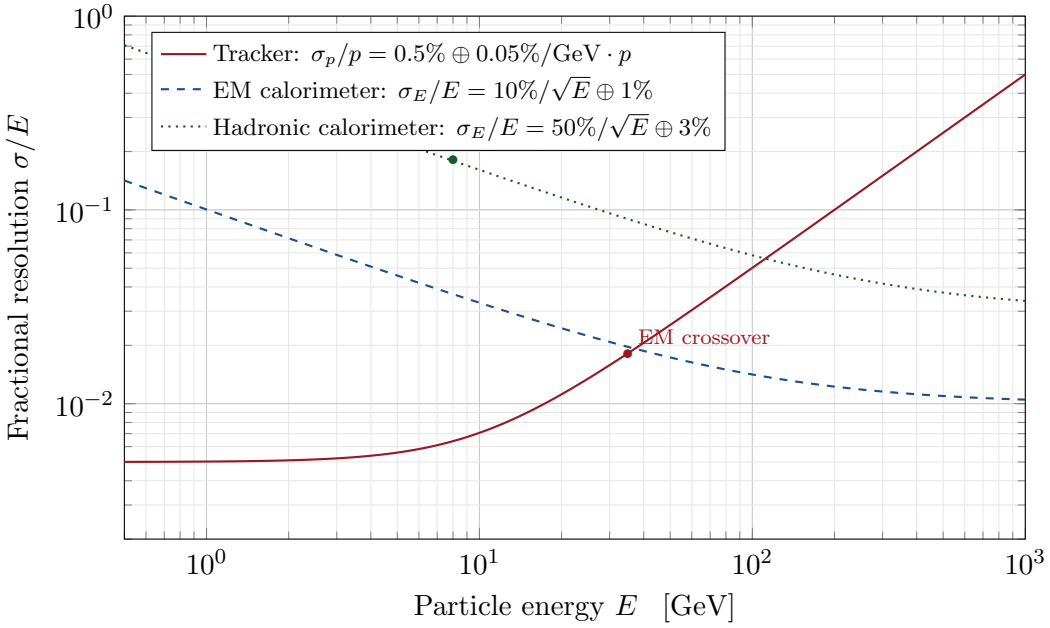
4 Step 4 – Comparing the two and solving for the crossover

Now we have both scalings explicitly:

$$\text{Tracker:} \qquad \left(\frac{\sigma_p}{p}\right)_{\text{trk}} = a_{\text{MS}} \oplus b\,p\,, \tag{16}$$

$$\text{Calorimeter:} \qquad \left(\frac{\sigma_E}{E}\right)_{\text{cal}} = \frac{a}{\sqrt{E/\text{GeV}}} \oplus b_{\text{cal}}\,. \tag{17}$$

4.1 Graphical comparison



At low energy, both calorimeter curves are sitting up near 10–50%, while the tracker is down at its multiple-scattering floor of a fraction of a percent – the tracker is the clear winner. As energy grows,

the tracker curve rises linearly while the calorimeter curves fall as $1/\sqrt{E}$; they cross, and calorimetry takes over.

4.2 Solving for the crossover energy algebraically

In the regime where the stochastic term of the calorimeter and the high-p term of the tracker both dominate (i.e., away from the low-p MS floor and the high-E constant term), we set them equal:

$$b\,E \;=\; \frac{a}{\sqrt{E/\mathrm{GeV}}}. \tag{18}$$

Solving for E:

$$E^{3/2} \;=\; \frac{a}{b}\sqrt{\mathrm{GeV}} \qquad \Longrightarrow \qquad \boxed{E_\times \;=\; \left(\frac{a}{b}\right)^{2/3} \mathrm{GeV}^{1/3}.} \tag{19}$$

Plugging in representative LHC-era values:

Case	a (stochastic)	b (tracker slope)	$E_\times$
EM calo / silicon tracker	0.10	5×10^{-4}/GeV	$(200)^{2/3} \approx 34\,\mathrm{GeV}$
EM calo / gas tracker	0.10	2×10^{-3}/GeV	$(50)^{2/3} \approx 13\,\mathrm{GeV}$
HCAL / silicon tracker	0.50	5×10^{-4}/GeV	$(1000)^{2/3} \approx 100\,\mathrm{GeV}$
HCAL / gas tracker	0.50	2×10^{-3}/GeV	$(250)^{2/3} \approx 40\,\mathrm{GeV}$
HCAL / compact low-B tracker	0.60	5×10^{-3}/GeV	$(120)^{2/3} \approx 24\,\mathrm{GeV}$
HCAL / hadron tower-level	0.80	10^{-2}/GeV	$(80)^{2/3} \approx 18\,\mathrm{GeV}$

Where does "5–10 GeV" come from?

The specific figure of merit 5–10 GeV quoted in the problem is the crossover for a charged hadron compared between (i) a modest, non-LHC-class tracker with weak B field and/or short lever arm, and (ii) a typical sampling hadron calorimeter. In a compact collider experiment (short L, modest B, $b \sim 10^{-2}$/GeV) the tracker degrades rapidly enough that even a mediocre hadronic calorimeter beats it by $E \sim 10\,\mathrm{GeV}$. At the high-precision LHC trackers, the crossover is pushed out to many tens of GeV; the problem's "5–10 GeV" value corresponds to a more typical collider, not the best-in-class ATLAS/CMS silicon systems. In *all* cases, the basic message is the same: there is some energy scale above which calorimetry is more precise, and below which tracking is.

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5 Step 5 – The deeper reason the two scalings go opposite ways

The derivations above show *what* the scalings are. The conceptual question is *why* they are opposite. The short answer is:

Geometry vs. statistics

A **tracker** measures a small displacement (the sagitta) and infers momentum from it. The displacement is what we *want* to be large and easy to see, and it shrinks as $1/p$ because fast particles bend less. So the *signal* shrinks with p, while the noise floor (position resolution, multiple scattering) stays fixed. Signal/noise degrades as p, so $\sigma_p/p \propto p$.

A **calorimeter** measures a sum of many independent shower quanta. The number of quanta *grows linearly* with energy, so the signal grows with E, while the relative fluctuation of a sum of N things scales as $\sqrt{N}$, giving a signal/noise that grows as $\sqrt{E}$. So $\sigma_E/E \propto 1/\sqrt{E}$.
At low energy the tracker signal is large and easy to measure; at high energy the calorimeter signal is large and easy to measure. Each instrument excels in the regime where *its* signal is biggest relative to *its* noise.

There is a useful analogy: the tracker is like trying to measure the deflection of a marble rolling through a weak wind (bigger momentum ⇒ tighter trajectory ⇒ deflection vanishes), while the calorimeter is like weighing a pile of sand by counting the grains (bigger pile ⇒ more grains ⇒ $\sqrt{N}$ Poisson noise drops relative to the total count). These are not the same measurement in disguise; they respond to energy inversely.

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6 Step 6 – Beyond precision: three more reasons calorimetry dominates at high energy

Precision alone would be enough justification, but real collider experiments have additional reasons to rely on calorimetry at high energy:

Three more reasons calorimetry dominates

1. **Neutrals are invisible to trackers.** Photons, neutrons, K_L^0's, Λ^0's, π^0's – none of these leave a track, because a tracker is blind to neutral particles. The only way to measure their energy is via calorimetry. For processes like $H \to \gamma\gamma$, $Z \to \nu\bar{\nu}$ (where missing transverse energy is crucial), or $B \to K^*\gamma$, the tracker is useless for the neutrals in the final state.
2. **Jets require calorimetry.** At $E \gtrsim 100\,\text{GeV}$, partons hadronize into collimated sprays of 20–50 particles covering both charged and neutral species. Summing calorimeter cells within a cone gives the jet energy in one shot; summing charged-track momenta misses the neutral fraction ($\sim$ 30% on average) and combinatorially mis-assigns tracks. The "particle-flow" approach used by CMS and ATLAS does combine both – but the calorimeter is the anchor.
3. **Speed.** Modern experiments collide at 25–40 MHz and must trigger within microseconds on which events to record. Calorimeter signals are fast (tens of ns), integrate cleanly, and can be read out in large contiguous regions for trigger decisions. Track reconstruction is slower and more combinatoric. Triggers at ATLAS/CMS are dominated by calorimeter sums at the first stage.

Summary: Full answer to Problem 5

Why precision flips at a few GeV

Tracker (momentum from B-field curvature). The sagitta of a curved track is $s = 0.3\,BL^2/(8p_T)$, which shrinks as $1/p_T$. Because the position resolution σ_x is fixed, the fractional momentum resolution grows linearly with momentum:

$$\frac{\sigma_{p_T}}{p_T} = a_{\text{MS}} \oplus b\,p_T, \qquad b \propto \frac{\sigma_x}{BL^2}.$$

Calorimeter (energy from shower counting). The shower size N is proportional to energy ($N \sim E/E_c$), and Poisson statistics on N independent samples give a $1/\sqrt{N}$ relative fluctuation. The fractional energy resolution therefore *shrinks* as $1/\sqrt{E}$:

$$\frac{\sigma_E}{E} = \frac{a}{\sqrt{E/\mathrm{GeV}}} \oplus b_{\mathrm{cal}}, \qquad a \sim 0.1 \text{ (EM)}, \ 0.5 \text{ (had)}.$$

Crossover. Equating the two stochastic/linear terms gives

$$E_\times = (a/b)^{2/3}\,\mathrm{GeV}^{1/3}.$$

This is $\sim$ 5–10 GeV for typical hadron/HCAL setups, $\sim$ 30–50 GeV for EM calorimeters with a good silicon tracker.

Conclusion. Below the crossover, the sagitta is large and the calorimeter shower statistics are few – tracking wins. Above the crossover, the sagitta shrinks below position resolution while the shower becomes statistically enormous – calorimetry wins. The fundamental reason is that tracking is a geometric measurement whose signal *shrinks* with energy while calorimetry is a counting measurement whose signal *grows* with energy; at enough energy, the latter always wins. Calorimetry additionally is the only way to measure neutrals, jets, and fast trigger primitives.